

Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
10.	(a)			student is correct because to find the minimum temperature they need ΔS and ΔH / they need to be able to calculate ΔG (1) they can calculate ΔH for both as standard enthalpy changes of elements in standard states are zero OR ΔH for decomposition is reverse of enthalpy change of formation (1) they can calculate ΔS for F_2O but not Cl_2O as we don't know entropy of Cl_2 OR entropy value of elements is not zero (1)	3			1 1 1		
	(b)	(i)		$2F_2O + Xe \rightarrow XeF_4 + O_2$	1	1				
		(ii)		xenon oxidation state changes from 0 to +4	1		1			
	(c)	(i)		$n(O_2) = \frac{pV}{RT} = \frac{1.01 \times 10^5 \times 82 \times 10^{-6}}{8.31 \times 288} = 3.46 \times 10^{-3} \text{ mol (1)}$ $[HF] = \frac{2 \times 3.46 \times 10^{-3}}{0.400} = 0.0173 \text{ mol dm}^{-3} \text{ (1)}$	2		1 1		1	
		(ii)		$K_a = \frac{[H^+][F^-]}{[HF]}$ so $[H^+]^2 = K_a \times [HF]$ (1) $[H^+]^2 = 6.6 \times 10^{-4} \times 0.0173 = 1.14 \times 10^{-5}$ so $[H^+] = 3.38 \times 10^{-3} \text{ mol dm}^{-3} \text{ (1)}$ $pH = 2.47 / 2.5 \text{ (1)}$ ecf possible from part (i)	3		1 1 1		1 1 1	

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	(d)	(i)		$K_c = \frac{[\text{HOCl}]^2}{[\text{Cl}_2\text{O}][\text{H}_2\text{O}]}$	1	1				
		(ii)	I	$[\text{H}_2\text{O}]$ at equilibrium = 0.29 mol dm ⁻³ (1) $[\text{Cl}_2\text{O}]$ at equilibrium = 0.03 mol dm ⁻³ (1) $K_c = 5.56$ (1) ecf possible from incorrect p expression	3		1 1	1	1 1 1	
			II	at a higher temperature K_c is larger so equilibrium has shifted to the right (1) reaction is endothermic (1) ecf possible from incorrect K_c value	2		1 1			
			III	Ti^+ and Fe^{2+} (1) SEP for HClO is more positive than for these half-cells so can oxidise the ions OR positive EMF values calculated for the reactions (1)	2		1 1			
					18	2	12	4	7	0